

DATA CLASSIFICATION IN SPSS: Grouped & Ungrouped Data + Levels of Analysis

Grouped vs Ungrouped Data

A. Ungrouped Data

- Data presented in **raw form**, individual values recorded as they are.
- No organization into classes or categories.
- Common at the **data entry** stage in SPSS.

Example (Ungrouped data in SPSS Data View):

Respondent	Age
1	21
2	24
3	19
4	35

Grouped Data

Grouped Data

- Data **organized into classes**, categories, or groups.
- Used for **summarizing** large datasets.

Example of grouped age categories:

Age Group	Frequency
18–25	45
26–35	38
36–45	22

In SPSS, grouping is done with:

Transform → Recode into Categories

or

Analyze → Descriptive Statistics → Frequencies → *Format into Groups*

Levels of Statistical Analysis

SPSS supports three main types:

A. Univariate Analysis

Analysis of **one variable** at a time

Used to measure **distribution, central tendency, dispersion**

Examples in SPSS:

- Frequencies → Frequencies / bar chart
- Descriptives → Mean, median, mode, std. dev.
- Explore → One variable analysis

Output Example

Mean Age = 26.4

Interpretation

Average respondent age

Bivariate Analysis

Analysis of **two variables together**
Used to examine **relationships**

Examples in SPSS:

Variable Type

Continuous vs Continuous

Categorical vs Categorical

Categorical vs Continuous

SPSS Procedure

Correlation → Pearson/Spearman, Scatterplot

Crosstab + Chi-square

T-test / ANOVA

Explore Command in SPSS

“**Explore**” is a powerful **univariate and bivariate diagnostic tool** found under:

Analyze → Descriptive Statistics → **Explore**

- What Explore Can Do

Output Feature

Mean, Median, Mode

Std Dev, Variance, Range

Boxplots

Normality Tests (Shapiro-Wilk /
Kolmogorov-Smirnov)

Confidence Intervals

Stem-and-Leaf / Histograms

Purpose

Measures of central tendency

Dispersion summary

Outlier detection

Check distribution type

Precision of estimates

Shape of data